Daksh Kohli

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Education

Bachelor of Engineering in Computer Engineering *2024 – 2028*
Thapar Institute of Engineering and Technology, Patiala CGPA: 8.39

Leadership & Experience

Technical Intern – TSLAS Office, TIET *2026 – Present*

- Co-developed an auto-scheduling engine as part of a 2-person team, generating timetables across 300+ major, minor, and elective combinations for institute-wide faculty use, under the Dean, TSLAS, using Python and simulated annealing-based optimization.
- Streamlined manual scheduling workflows across multiple departments, cutting redundant operational effort and freeing up team bandwidth for planning and execution initiatives.

Technical Secretary – Microsoft Learn Student Chapter (MLSC), TIET *2025 – Present*

- Lead technical initiatives and cross-functional collaboration across a 14-member executive board and 40+ contributors, driving execution on 5+ concurrent projects.
- Mentored 30+ students across 3 hackathons, and manage task delegation, sprint planning, and team communication across 4+ concurrent technical projects.

Projects

Constraint-Based Timetable Scheduling System – TSLAS Office, TIET
Python, Next.js, AWS, DynamoDB, Docker

- Engineered a production-grade timetable scheduling system for 800+ students and 30+ faculty members, deployed on AWS with DynamoDB, supporting 20+ configurable academic constraints.
- Implemented a simulated annealing optimization algorithm to resolve faculty overlaps, room-capacity limits, elective clashes, and cross-department timetable conflicts.
- Designed and integrated a CI/CD pipeline with 11 automated validation tests covering room, faculty, and student clashes to ensure timetable integrity before deployment.

Thapar Library Management System – Official Institute Project
Tech Lead
Next.js, FastAPI, PostgreSQL, Redis, Docker, Nginx

- Leading a 7-member development team as Tech Lead to rebuild Thapar Institute's official library management platform, targeting 10,000+ monthly active users.
- Architected the system with Redis-backed real-time updates and concurrency control to safely handle simultaneous book checkouts across 10,000+ monthly users, while owning sprint planning and code review as Tech Lead.

Personal Document Search Engine – Classical IR + ML-Ranked Hybrid Retrieval GitHub Link
Java, Spring Boot, React, PDFBox, Apache POI, Docker

- Built a full-text search engine from first principles: a custom tokenizer, Porter stemmer, and hand-built inverted index storing term positions to support phrase queries, rather than relying on a library like Lucene or Elasticsearch.
- Implemented TF-IDF and BM25 ranking algorithms manually, with cosine-similarity scoring as an alternate retrieval path, and benchmarked the two against each other on ranking quality.

Volatility Regime Forecasting with Deep Learning GitHub Link
Python, PyTorch, LSTM/Transformer, Scikit-learn

- Engineered a financial time-series pipeline across 9 liquid ETFs and labeled daily volatility regimes via GMM clustering (73% low / 25% medium / 1% high-vol), converting the problem into supervised classification.
- Benchmarked 4 architectures on regime classification – Gradient Boosted Trees reached 95.2% accuracy / 0.947 macro-F1, outperforming Logistic Regression (92.0%), LSTM (85.7%), and Transformer (85.0%) – then backed the top model with a simple exposure rule that beat buy-and-hold on Sharpe ratio (0.98 vs. 0.91) despite lower raw return.

Technical Skills

Languages: Python, Java, C++, JavaScript
Backend & APIs: FastAPI, Spring Boot, Node.js, Express.js, REST API Design, RESTful Services
Frontend: Next.js, React.js, Tailwind CSS, Framer Motion, Three.js, React Three Fiber
Databases: PostgreSQL, MySQL, MongoDB, Redis
DevOps & Infrastructure: AWS, DynamoDB, Docker, Nginx, Cloudflare, Railway, CI/CD, Git, GitHub, Postman
Machine Learning: PyTorch, LSTM, Transformers, Scikit-learn, GMM Clustering, Learning-to-Rank, TF-IDF/BM25, Semantic Search
Auth & Platform Services: Clerk
Core Concepts: Database Management Systems (DBMS), Object-Oriented Programming (OOP), System Design, Data Modeling, Transaction Management, Concurrency Control, API Design
Project Management & Collaboration: SCRUM, JIRA, Agile Workflow, Technical Team Leadership, Stakeholder Communication, Cross-Functional Collaboration, Mentorship